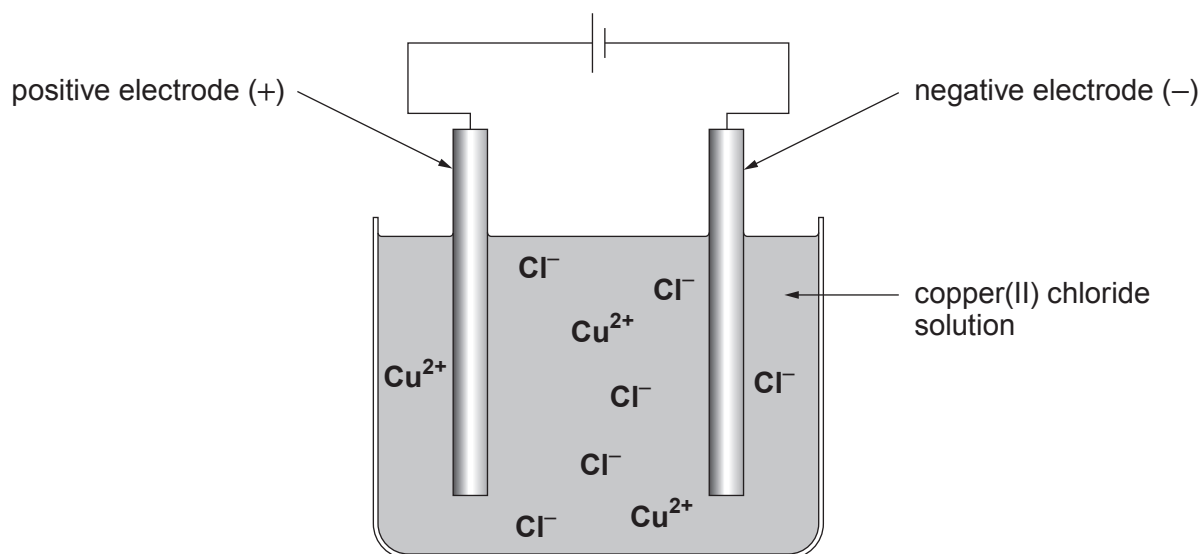


2. (a) Corey and Bailey used the following apparatus to obtain copper from copper(II) chloride solution.



(i)

electrolysis	copper	anode
cathode	chlorine	displacement
copper(II) chloride	neutralisation	

Choose words from the box to complete the following sentences.

[3]

The electrolyte used in this process is

Copper is formed at the negative electrode. This electrode is called the

.....

The breaking down of a compound using an electric current is called

.....



- (ii) Copper(II) chloride contains the ions Cu^{2+} and Cl^- .

Give the formula of copper(II) chloride.

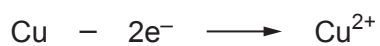
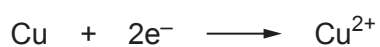
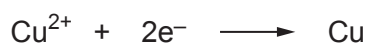
[1]

.....

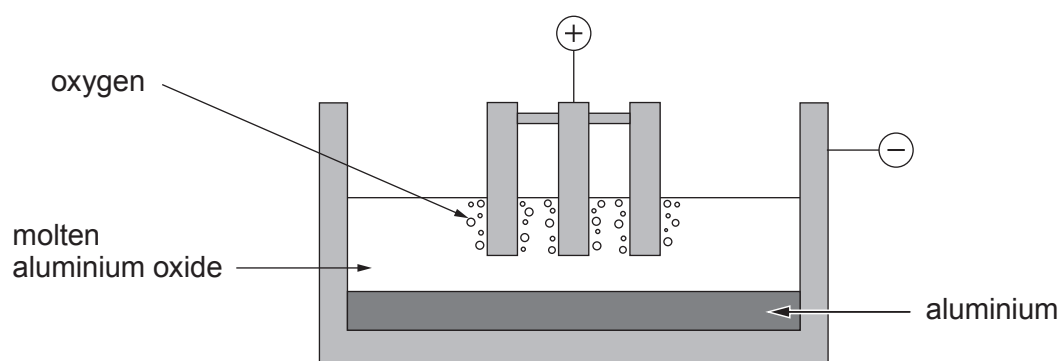
- (iii) During the process, copper ions become copper atoms by gaining electrons.

Tick (✓) the box next to the equation that shows this.

[1]

☐☐☐☐

- (b) The diagram represents the industrial extraction of aluminium from its ore.



- (i) Give the reason why the aluminium oxide must be molten during the extraction.

Tick (✓) the correct box.

[1]

to release the oxygen gas from the aluminium oxide

☐

to allow the aluminium ions and oxide ions to move

☐

to allow the aluminium to leave the cell

☐

to speed up the process

☐

- (ii) Put a number in the box to balance the equation that represents the overall reaction taking place.

[1]



- (iii) Tick (✓) the boxes next to the **two** factors that are important when choosing the location for aluminium plants in the UK. [2]

Aluminium plants in the UK should be close to

landfill sites

☐

power stations

☐

limestone quarries

☐

oil refineries

☐

coastal ports

☐

coal mines

☐

- (c) Aluminium is commonly used to make overhead power cables.



Other than being a good electrical conductor, give **two** properties which make aluminium a suitable material for making overhead power cables. [2]

Property 1

Property 2

